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2 Vague but exciting



Data Rep. in RDBMS

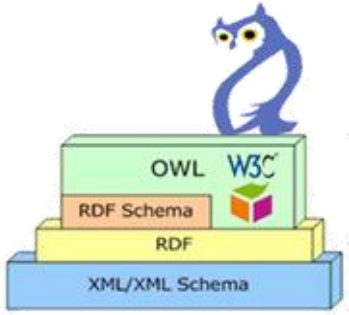
Real World Entity

Conceptual Level | Entity-Relationship Model

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep. in GGG

Real World Entity

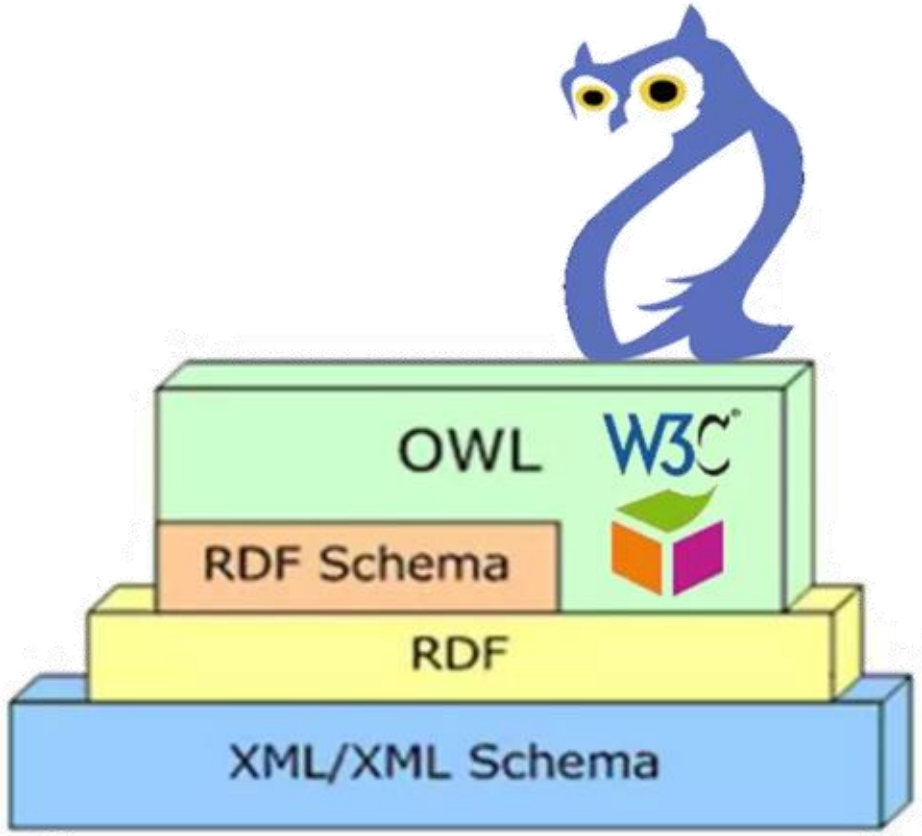
Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

A Data Odyssey: Web Ontology Language (OWL)



From RDF to RDFS to OWL: The Semantic Web Stack Evolution



Outline: W08: Semantic Web Modeling



I) XML: eXtendable Markup Language

II) RDF: Resource Description Framework

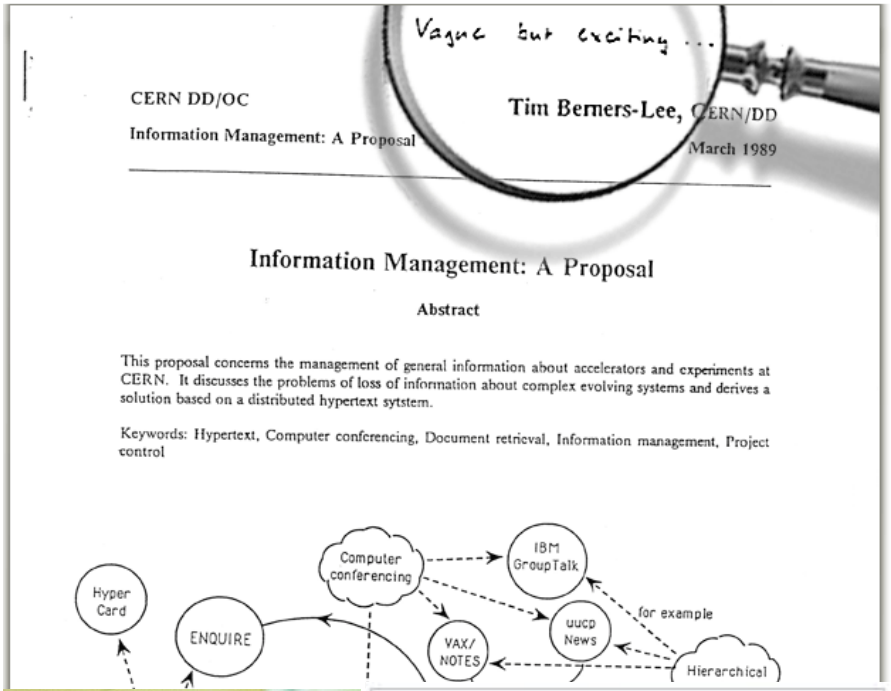
III) RDFS: RDF Schema

IIII) OWL: Web Ontology Language

Hands-On

1989: World Wide Web (WWW, The Web)

https://www.w3.org/Proposal.html
https://www.w3.org/History/1989/proposal.html
Conseil Européen pour la Recherche Nucléaire (CERN in French)
European Laboratory for Particle Physics
https://home.cern/


Vague but exciting ...

CERN DD/OC

Tim Berners-Lee, CERN/DD

Information Management: A Proposal

March 1989

Information Management: A Proposal

Abstract

This proposal concerns the management of general information about accelerators and experiments at CERN. It discusses the problems of loss of information about complex evolving systems and derives a solution based on a distributed hypertext sytstem.

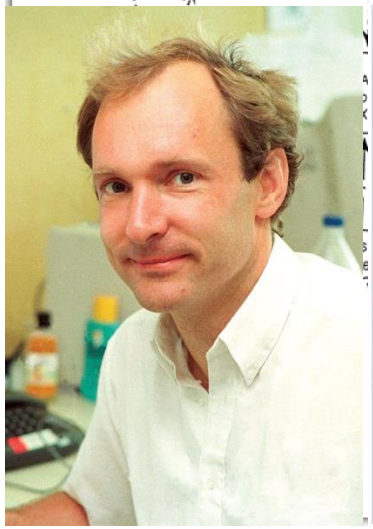
Keywords: Hypertext, Computer conferencing, Document retrieval, Information management, Project control

For human-consumption publicly shared

Information

Knowledge

The Web Technologies: HTML, URL, HTTP, …





This NeXT Computer was used by Berners-Lee at CERN and became the world's first web server.

Sir Timothy John Berners-Lee (1955-Present), aka TimBL, English Computer Scientist
Professorial Research Fellow, University of Oxford, Professor Emeritus at MIT

1989: HTML: Markup Language

Inception is a 2010 science fiction action and adventure film directed by Christopher Nolan. The film is 148 minutes long. Christopher Nolan (born 30 July 1970) is an English filmmaker.

- Traditional Publishers (pre-computer era): editors used red pens to mark font styles, sizes, and layout instructions on physical proofs.
- Generalized Markup Language (GML, 1969): at IBM to automate document handling and legal paperwork, laying the foundation for computer-based structure.
- Standard Generalized Markup Language (SGML, 1986): from GML into an international standard (ISO 8879) for defining device-independent, platform-independent document structures.
- HyperText Markup Language (HTML, 1989): Invented by Tim Berners-Lee as a simple SGML subset to format text and link scientific documents across the early Web.

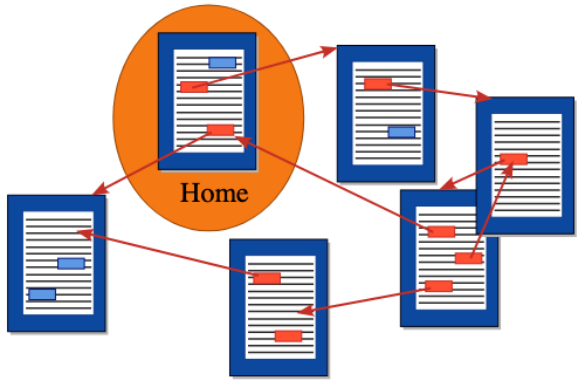
1989: HTML: Markup Language

Inception is a 2010 science fiction action and adventure film directed by Christopher Nolan. The film is 148 minutes long. Christopher Nolan (born 30 July 1970) is an English filmmaker.

```
<p><span style="color:yellow;">Inception</span> is a <span
style="color:cyan;">2010</span><span style="color:green;">science fiction,
action, adventure</span> film directed by <span
style="color:gray;">Christopher Nolan</span>. The film is <span
style="color: purple;">148</span> minutes long. <span
style="color:gray;">Christopher Nolan</span> (born <span
style="color:pink;">30 July 1970</span>) is an <span
style="color:orange;">English</span> filmmaker.</p>
```

1989: HTML: HyperText Markup Language

'Hypertext' is a recent coinage. 'Hyper-' is used in the mathematical sense of extension and generality (as in 'hyperspace,' 'hypercube') rather than the medical sense of 'excessive' ('hyperactivity'). There is no implication about size, a hypertext could contain only 500 words or so. 'Hyper-' refers to structure and not size."
—Theodor H. Nelson, Brief Words on the Hypertext, 1967




```
<a href="http://fani-lab.github.io/index.html">Visit our lab!</a>
```

- Annotating (tagging) text, i.e., adding meta data (e.g., color) about data.

```
<span style="color:yellow;">Inception</span>
```

- Syntactically distinguishable from the actual text, so **machine-readable**.
- **World Wide Web Consortium (W3C)** standardizes so all web browsers have same interpretation.

1989: HTML: Not eXtendable Unless by W3C

Inception is a 2010 science fiction action and adventure film directed by Christopher Nolan. The film is 148 minutes long. Christopher Nolan (born 30 July 1970) is an English filmmaker.

```
<p><span entity="movie">Inception</span> is a <span
year="release">2010</span><span type="genre">science fiction, action,
adventure</span> film directed by <span entity="director">Christopher
Nolan</span>. The film is <span time="duration">148</span> minutes long.
<span entity="person">Christopher Nolan</span> (born <span
datetime=”birthdate">30 July 1970</span>) is an <span
location="country">English</span> filmmaker.</p>
```



1996: XML: eXtendable Markup Language

Can be extended to new user-defined tags:

```
<Movie>
        <Id>1</Id>
        <Title>Inception</Title>
        <Genre>Adventure</Genre>
        <Genre>Sci-fi</Genre>
        <RunningTime>148</RunningTime>
        <ReleaseDate>2010-07-08</ReleaseDate>
        <DirectorId>1</DirectorId>
</Movie>
<Director>
        <Id>1</Id>
        <FirstName>Christopher</FirstName>
        <LastName>Nolan</LastName>
        <BirthDate>1970-06-30</BirthDate>
        <BirthPlace>England</BirthPlace>
</Director>
```

1996: XML: eXtendable Markup Language

- 1996: Officially proposed and work began by **W3C's XML Working Group**.
- 1998: W3C released **XML 1.0** as an official W3C recommendation.
- 2006: **XML 1.1** was published by the W3C



XML: Why?

Files are **self-describing** due to presence of tags, like a **document**!

→ Both **human-** and **machine-**readable.

→ *de facto* for **small-scale** data exchange:

- Person to Person
- Person to Software (upload)
- Software to Person (download)
- Software to Software (Browser, Web Services, APIs, …)



Relational to XML vs. CSV

Movie			
Id	Title	RunningTime	ReleaseDate
1	Inception	148	2010

MovieGenre	
MovieId	GenreId
1	6
1	1

MovieDirector	
MovieId	DirectorId
1	1

Genre	
Id	Title
1	Sci-fi
2	Drama
3	Crime
4	Mystery
5	Thriller
6	Adventure
7	Horror
8	Action

Director				
Id	FirstName	LastName	DateOfBirth	PlaceOfBirth
1	Christopher	Nolan	1970-06-30	England

```
SELECT ...
FROM Movie, MovieGenre, Genre
WHERE  Movie.Title = 'Inception'
   AND Movie.Id = MovieGenre.MovieId
   AND MovieGenre.GenreId = Genre.Id
```

```
SELECT ...
FROM Movie, MovieDirector, Director
WHERE  Movie.Title = 'Inception'
   AND Movie.Id = MovieDirector.MovieId
   AND MovieDirector.DirectorId = Director.Id
```

Serializing (generator) …

Relational to XML vs. CSV

```xml
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
        <Movie>
                <Id>1</Id>
                <Title>Inception</Title>
                <Genre>Adventure</Genre>
                <Genre>Sci-fi</Genre>
                <RunningTime>148</RunningTime>
                <ReleaseDate>2010-07-08</ReleaseDate>
                <DirectorId>1</DirectorId>
        </Movie>
        <Director>
                <Id>1</Id>
                <FirstName>Christopher</FirstName>
                <LastName>Nolan</LastName>
                <BirthDate>1970-06-30</BirthPlace>
                <BirthDate>England</BirthDate>
        </Director>
</MoviesionData>
```

XML Serializer (Generator):

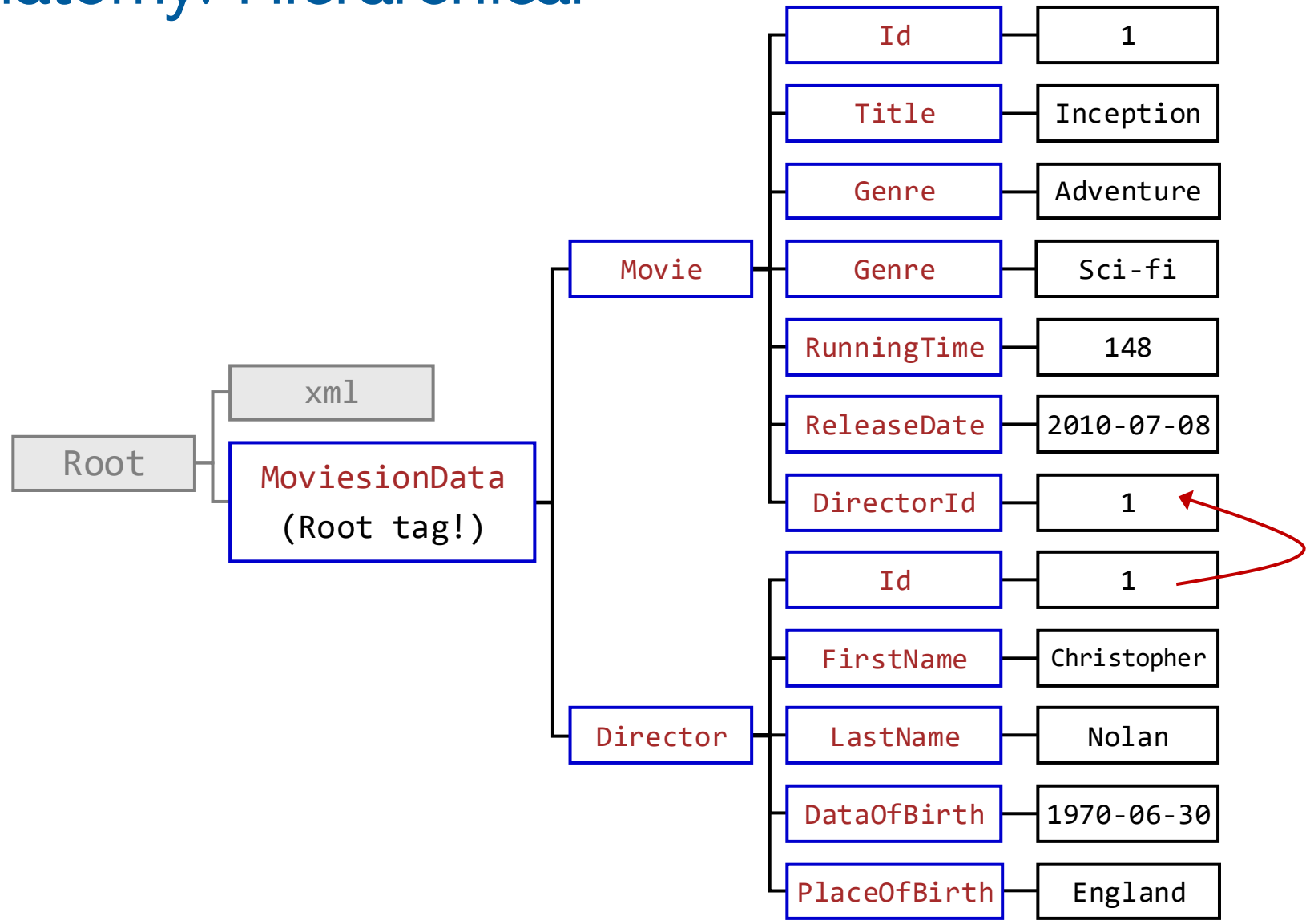
+ Self-explanatory
– Long document!

```
1,Inception,Adventure,Sci-fi,148,2010-07-08,1
1,Christopher,Nolan,1970-06-30,England
```

CSV Serializer (Generator):

+ Short document
– No meta-data, no or mix semantics!

XML: Anatomy: Hierarchical



XML: Anatomy: Element

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
	<Movie>
		<Id>1</Id>
		<Title>Inception</Title>
		<Genre>Adventure</Genre>
		<Genre>Sci-fi</Genre>
		<RunningTime>148</RunningTime>
		<ReleaseDate>2010-07-08</ReleaseDate>
		<DirectorId>1</DirectorId>
	</Movie>
	<Director>
		<Id>1</Id>
		<FirstName>Christopher</FirstName>
		<LastName>Nolan</LastName>
		<BirthDate>1970-06-30</BirthDate>
		<BirthPlace>England</BirthPlace>
	</Director>
</MoviesionData>
```

Element 01

XML: Anatomy: Element

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
        <Movie>
                <Id>1</Id>
                <Title>Inception</Title>
                <Genre>Adventure</Genre>
                <Genre>Sci-fi</Genre>
                <RunningTime>148</RunningTime>
                <ReleaseDate>2010-07-08</ReleaseDate>
                <DirectorId>1</DirectorId>
        </Movie>
        <Director>
                <Id>1</Id>
                <FirstName>Christopher</FirstName>
                <LastName>Nolan</LastName>
                <BirthDate>1970-06-30</BirthDate>
                <BirthPlace>England</BirthPlace>
        </Director>
</MoviesionData>
```

Element 02

Element 03

XML: Anatomy: Element

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
        <Movie>
                <Id>1</Id>                                  Element 04
                <Title>Inception</Title>                    Element 05
                <Genre>Adventure</Genre>                    Element 06
                <Genre>Sci-fi</Genre>                       Element 07
                <RunningTime>148</RunningTime>              Element 08
                <ReleaseDate>2010-07-08</ReleaseDate>       Element 09
                <DirectorId>1</DirectorId>                  Element 10
        </Movie>
        <Director>
                <Id>1</Id>                                  Element 11
                <FirstName>Christopher</FirstName>          Element 12
                <LastName>Nolan</LastName>                  Element 13
                <BirthDate>1970-06-30</BirthDate>           Element 14
                <BirthPlace>England</BirthPlace>            Element 15
        </Director>
</MoviesionData>
```

XML: Anatomy: Element: Rules

- Made up of an opening tag and an ending tag with same name

```
<Title>Inception</Title>
```

- Properly nested elements

```
<Movie>
        <Id>1</Id>
        <Title>Inception</Title>
</Movie>
```

```
<Movie>
        <Id>1</Id>
        <Title>Inception
</Movie></Title>
```

Improperly nested → Not well-formed!

XML: Anatomy: Element: Rules

- Tags are case-sensitive `<Title></Title>` ≠ `<title></title>`
- Empty elements can be abbreviated `<Title></Title>` ⟶ `<Title/>`
- Multiple elements may have same name

```
<Movie>
        <Genre>Adventure</Genre>
        <Genre>Sci-fi</Genre>
</Movie>
```

- Order matters

```
<Movie>
        <Genre>Adventure</Genre>
        <Genre>Sci-fi</Genre>
</Movie>
```

≠

```
<Movie>
        <Genre>Sci-fi</Genre>
        <Genre>Adventure</Genre>
</Movie>
```

XML: Anatomy: Element with Attributes

- An element may have attributes key="value" in opening tag.
- Key must be unique within the element.
- Attributes cannot be nested.

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
          <Movie>
                    <Id>1</Id>
                    <Title>Inception</Title>
                    <Genre>Adventure</Genre>
                    <Genre>Sci-fi</Genre>
                    <RunningTime>148</RunningTime>
                    <ReleaseDate>2010-07-08</ReleaseDate>
                    <DirectorId>1</DirectorId>
          </Movie>
          <Director>
                    <Id>1</Id>
                    <FirstName>Christopher</FirstName>
                    <LastName>Nolan</LastName>
                    <BirthDate>1970-06-30</BirthDate>
                    <BirthPlace>England</BirthPlace>
          </Director>
</MoviesionData>
```

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
          <Movie Id="1" Title="Inception">
                    <Genre>Adventure</Genre>
                    <Genre>Sci-fi</Genre>
                    <RunningTime>148</RunningTime>
                    <ReleaseDate>2010-07-08</ReleaseDate>
                    <DirectorId>1</DirectorId>
          </Movie>
          <Director Id="1" BirthPlace="England">
                    <FirstName>Christopher</FirstName>
                    <LastName>Nolan</LastName>
                    <BirthDate>1970-06-30</BirthDate>
          </Director>
</MoviesionData>
```

XML: Anatomy: Element with Attributes

- An element may have attributes key="value" in opening tag.
- Key must be unique within the element.
- Attributes cannot be nested.

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
          <Movie Id="1" Title="Inception" RunningTime="148" ReleaseDate="2010-07-08" DirectorId=1>
                    <Genre>Adventure</Genre>
                    <Genre>Sci-fi</Genre>
          </Movie>
          <Director Id="1" BirthPlace="England" FirstName="Christopher" LastName="Nolan" BirthDate="1970-06-30"</Director>
</MoviesionData>
```

XML: Anatomy: Element vs. Attributes

Attributes are part of markup (tag) but elements are for the data (contents) …

So, representing data as attributes is not encouraged.

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
        <Movie style="color:blue">
                <Id>1</Id>
                <Title style="font:bold">Inception</Title>
                <Genre>Adventure</Genre>
                <Genre>Sci-fi</Genre>
                <RunningTime>148</RunningTime>
                <ReleaseDate>2010-07-08</ReleaseDate>
                <DirectorId>1</DirectorId>
        </Movie>
        <Director>
                <Id>1</Id>
                <FirstName>Christopher</FirstName>
                <LastName>Nolan</LastName>
                <BirthDate>1970-06-30</BirthDate>
                <BirthPlace>England</BirthPlace>
        </Director>
</MoviesionData>
```

XML: Anatomy: Element vs. Attributes

Attributes are part of markup (tag) but elements are for the data (contents) …

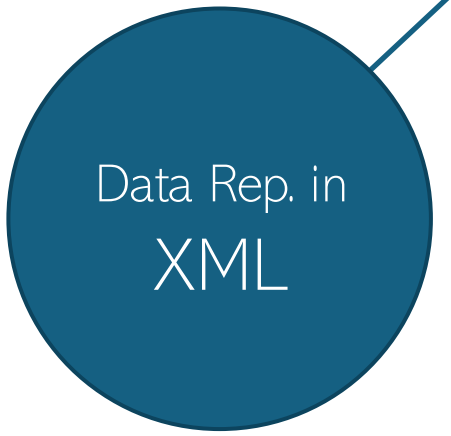
So, representing data as attributes is not encouraged.

Except PK or FK elements like Id, to uniquely access the element.

```
<?xml version="1.0" encoding="UTF-8"?>
<MoviesionData>
        <Movie Id="1" DirectorId="1" style="color:blue">
                <Title>Inception</Title>
                <Genre>Adventure</Genre>
                <Genre>Sci-fi</Genre>
                <RunningTime>148</RunningTime>
                <ReleaseDate>2010-07-08</ReleaseDate>
        </Movie>
        <Director Id="1">
                <FirstName>Christopher</FirstName>
                <LastName>Nolan</LastName>
                <BirthDate>1970-06-30</BirthDate>
                <BirthPlace>England</BirthPlace>
        </Director>
</MoviesionData>
```

XML: Anatomy: Hierarchical: Elements: Attributes

https://www.liquid-technologies.com/online-xml-validator



Real World Entity

Conceptual Level | Entity-Relationship Model

Logical Level | XML Schema Definition (XSD)

Physical Level | **File: XML Document**

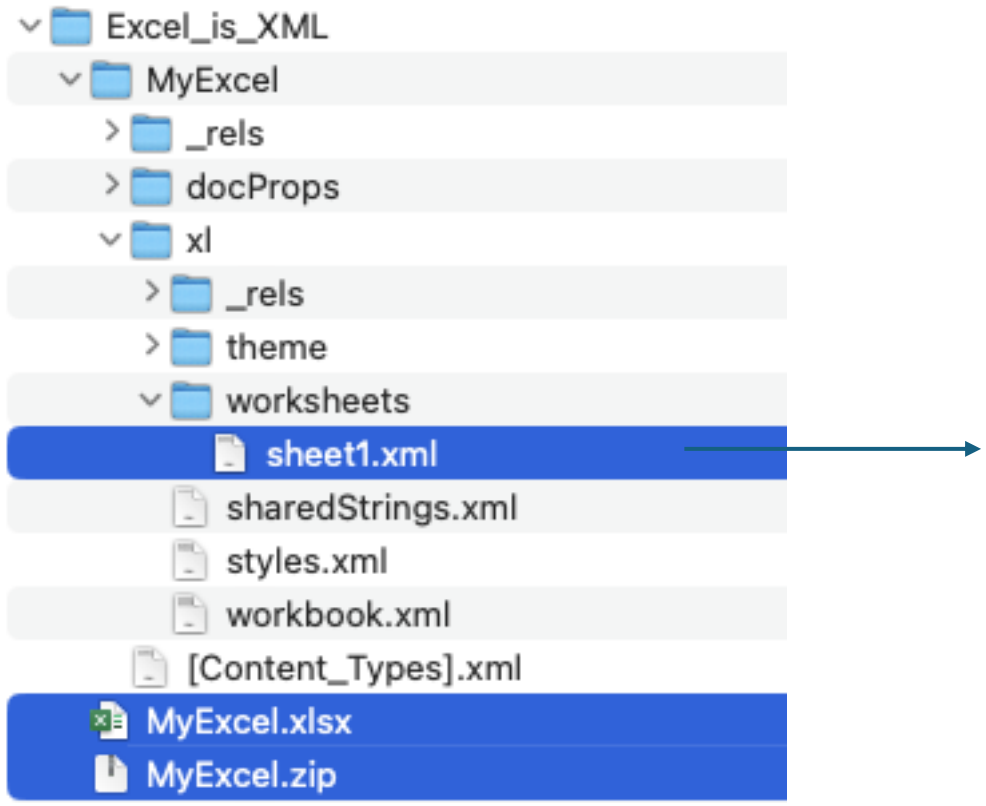
Computable Entity



Well-formed

Excel is XML (File → Document)

XML is the backbone and underlying format of Excel files (documents)



https://xml.onlineviewer.net/

